EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

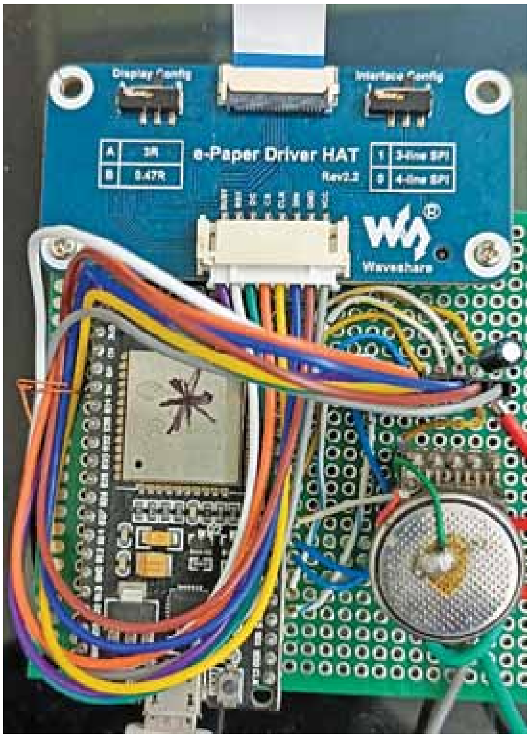

Fig. 4: The wired circuit board

to I2C pins as per circuit diagram.

For 3.3 volts, a low-power, low-drop rate HT7333A voltage regulator has been used, whose quiescent current is 4 microamperes. The input voltage is 12V to 3.47V, ensuring that when the input voltage reduces to 3.47V, you still get 3.3V for running the clock.

The power consumption of the clock is about 60 milliamperes on a 3.7-volt cell. On a very old (18650) LiPo cell, the clock runs nonstop for 2.5 days. The clock is best suited for solar-driven hall clocks, street clocks, or park clocks where a couple of days of non-availability of sunlight will not stop the operations. The e-paper viewing angle is >170 degrees and the display is visible even under broad daylight. However, during a dark night or in the absence of light, e-paper will not be visible. It needs incident light to work; but e-paper does not have any light! In case a solar panel is used, then the charging module needs to be connected, which is not provided in the schematic.

Just power the device and your personalised e-paper clock is ready. You can see the clock and the calendar on the display in real time. EFY

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